TRACKPOINT MANAGEMENT

Project Description

TrackPoint Management is a location-tracking application designed to track live GPS coordinates in real time and provide visual representations of device locations on a map. The system is intended as a practical tool for seminars, educational environments, and ethical hacking demonstrations where location-sharing, permission handling, and map-based visualization can be demonstrated in a controlled and authorized environment.

Problem Statement

Tracking information is often difficult to demonstrate as a complete workflow because device status, location coordinates, sharing links, consent, and map visualization may exist in separate interfaces. TrackPoint brings these activities together so an authorized user can manage devices, generate a tracking link, request access, and visualize the resulting GPS location on a map.

Why does this problem exist?

- Live GPS coordinates need to be collected and displayed in a clear, understandable workflow.
- Device information and location status can be difficult to inspect from a single interface.
- Sharing a tracking link requires a clear permission/consent step before location information is exposed.
- Map visualization is needed to turn latitude and longitude into an understandable geographic representation.
- Seminars and educational demonstrations benefit from a simple end-to-end example of location tracking and access control.

Stakeholders

Internal users:

- Project Developer / Application Developer
- Data / Backend Engineer
- Dashboard / UI Designer
- System Administrator

External / demonstration users:

- Authorized application users
- Seminar and classroom participants
- Cybersecurity / ethical hacking learners
- Application testers and demonstrators

Expected system users:

- Registered users who manage their own devices
- Authorized recipients who open a shared tracking link
- Administrators responsible for users, devices, links and settings

Objectives, Outputs and User Activity

Objectives of the Project

1. Provide real-time tracking of GPS coordinates for authorized devices.

2. Present device locations visually on a map.

3. Allow users to manage and inspect their registered devices.

4. Generate tracking links that can be shared with another user.

5. Require permission / consent before revealing a target device's location.

6. Display device name and basic information along with latitude, longitude and update information.

7. Integrate map visualization through the Google Street Map / Google Maps API as configured by the application.

8. Provide an administrator area for user, device, link and settings management.

9. Make the system suitable for seminars, educational environments and controlled ethical hacking demonstrations.

10. Provide a clear, understandable end-to-end tracking workflow.

Expected Outputs

- A working TrackPoint application.
- Live or latest available GPS coordinates for authorized devices.
- Device list with online/offline status and device details.
- Generated tracking links for controlled sharing.
- Permission / consent result before location disclosure.
- Map-based visualization of device location.
- Administrative management functions.
- A report that can document the demonstration and workflow.

User Activity

- Register or log in to the application.
- Open the dashboard.
- View My Devices and device status.
- View device details.
- Create a tracking link.
- Copy or share the link with another authorized user.
- Open the map and view the device location when access is permitted.
- Request or provide permission / consent.
- View device name and basic information.
- View latitude and longitude and the latest update information.
- Download/share a report where supported.
- Give feedback and log out.

Users, Technologies and Methodology

User / Tracker Functions

- Register / Login
- View My Devices
- View list of devices with online/offline status
- View device details
- Create a tracking link
- Generate and copy a tracking link
- Send the link to another user
- Open the map
- View live/latest device location
- Request permission / consent
- View device name and basic information after access is allowed
- View location on Google Street Map / configured Google Maps integration
- Download / export a report where implemented
- Give feedback
- Logout

Admin / Data Engineer Functions

- Login
- Manage users
- Manage devices
- View all tracking links
- Monitor GPS/location data
- Manage permissions and access
- Manage application settings
- Review feedback
- Export or back up data where supported
- Logout

Technologies / Prerequisites

- **Node.js v16 or higher** — required runtime for the application.
- **npm** — used to install project dependencies and run the application scripts.
- **TrackPoint project directory** — the application is started from the **trackpoint** directory.
- **Map API integration** — used for visual map representation according to the application's configured API.
- **GPS/location source** — provides the coordinates that the application displays.

Installation and Run Commands

```
cd trackpoint
npm install
npm start
```

Login Credentials for the supplied demonstration configuration

```
username: admin
password: admin
```

Methodology

1. Application setup and dependency installation.

2. User authentication and dashboard access.

3. Device registration and device-status management.

4. GPS coordinate collection / retrieval.

5. Tracking-link generation and controlled sharing.

6. Permission / consent verification.

7. Location visualization on the map.

8. Administrative monitoring and management.

9. Testing of the complete user-to-map workflow.

10. Validation of access control and demonstration behavior.

Functional and Non-Functional Requirements

Functional Requirements

1. The system shall allow a user to register and log in.

2. The system shall authenticate users before providing protected dashboard functions.

3. The system shall allow users to view their registered devices.

4. The system shall show device online/offline status where available.

5. The system shall allow users to view device details.

6. The system shall generate a tracking link for a selected device / target workflow.

7. The system shall allow the generated link to be copied or shared with another user.

8. The system shall request permission / consent before revealing the target location.

9. If permission is denied, the system shall deny access and return the user to the appropriate sharing workflow.

10. If permission is allowed, the system shall show the device name and basic information.

11. The system shall show the device location using latitude and longitude.

12. The system shall display the location visually using the configured Google Street Map / Google Maps integration.

13. The system shall provide an administrator interface for user and device management.

14. The system shall allow administrators to view tracking links and manage application settings.

15. The system shall support feedback and logout.

Non-Functional Requirements

1. The application should respond to normal dashboard actions promptly.

2. GPS information shown to the user should be accurate to the available device/location source.

3. The application should protect location information behind authentication and permission/consent controls.

4. The interface should be simple and understandable for demonstrations and educational use.

5. The system should support multiple devices and users without unnecessary performance degradation.

6. The application should be maintainable so future GPS/map functionality can be added or updated.

7. The system should work in an environment capable of running Node.js v16 or higher.

8. Location and user data should be handled securely and access should be limited to authorized users.

Ethical Measures, Privacy and Security

Ethical Use

- TrackPoint should be demonstrated only with authorized devices and participants.
- Location tracking must not be used to secretly monitor or stalk another person.
- Ethical hacking demonstrations should use explicit authorization and controlled test environments.
- The tracking-link workflow should make permission / consent an explicit part of location access.
- Demonstration coordinates and sample devices should be used when real personal locations are not necessary.

Permission and Consent

- The target user clicks the tracking link.
- The application presents a permission / consent step.
- When access is denied, location information is not disclosed.
- When access is allowed, the application can show the permitted device information and location.
- The consent flow should be clearly explained during seminars and demonstrations.

Data Privacy and Security

- Collect only the location and device information necessary for the application's purpose.
- Do not expose private information unnecessarily.
- Protect authenticated routes and administrative functions.
- Restrict location information to authorized users and permitted sharing workflows.
- Avoid storing historical location data unless it is genuinely required by the application.
- Secure any database, logs, API credentials and configuration values.
- Use HTTPS and secure deployment practices when the application is exposed beyond a local demonstration environment.

Demonstration Safety

- Use test devices or consenting participants.
- Do not use the system to obtain the location of a person without authorization.
- Do not publish real people's coordinates in screenshots, reports or seminars.
- Use fictional/sample coordinates for documentation whenever possible.
- Clearly distinguish a controlled ethical hacking demonstration from unauthorized tracking.

Project Limitation

The supplied project requirements establish the tracking, mapping, consent and administrative workflow. They do not specify the exact GPS hardware implementation, database technology, authentication framework, API key configuration, or historical-location retention model. Those implementation details should be documented separately if they exist in the actual source code.

Project Purpose

TrackPoint is positioned as a practical educational and demonstration application for understanding GPS data, map visualization, access control, permission handling and responsible cybersecurity testing.

Figure 1 — Use Case Diagram: User / Tracker

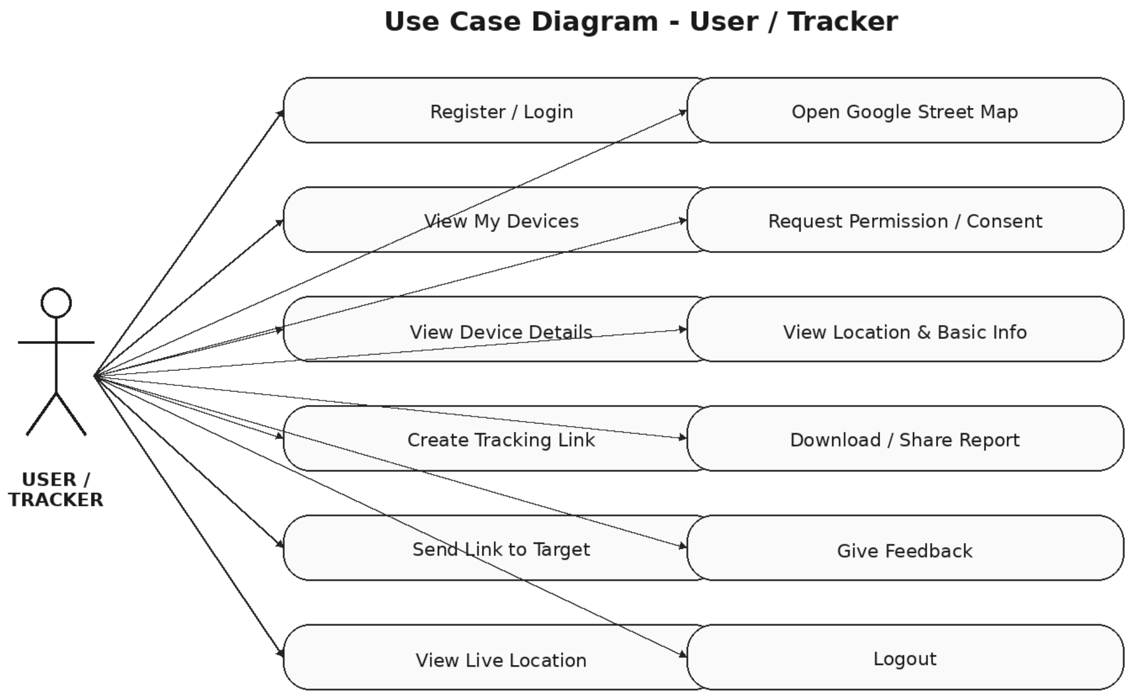

Figure 1: Use case diagram showing the interactions available to the User / Tracker actor in TrackPoint Management.

Diagram interpretation: The user authenticates, manages devices, creates and shares a tracking link, accesses the map, requests or receives consent, views permitted location information, and can provide feedback or log out.

Key workflow: Login → Dashboard → Device / Link / Map functions → Permission / Consent → Authorized Location View.

Figure 2 — Use Case Diagram: Admin / Data Engineer

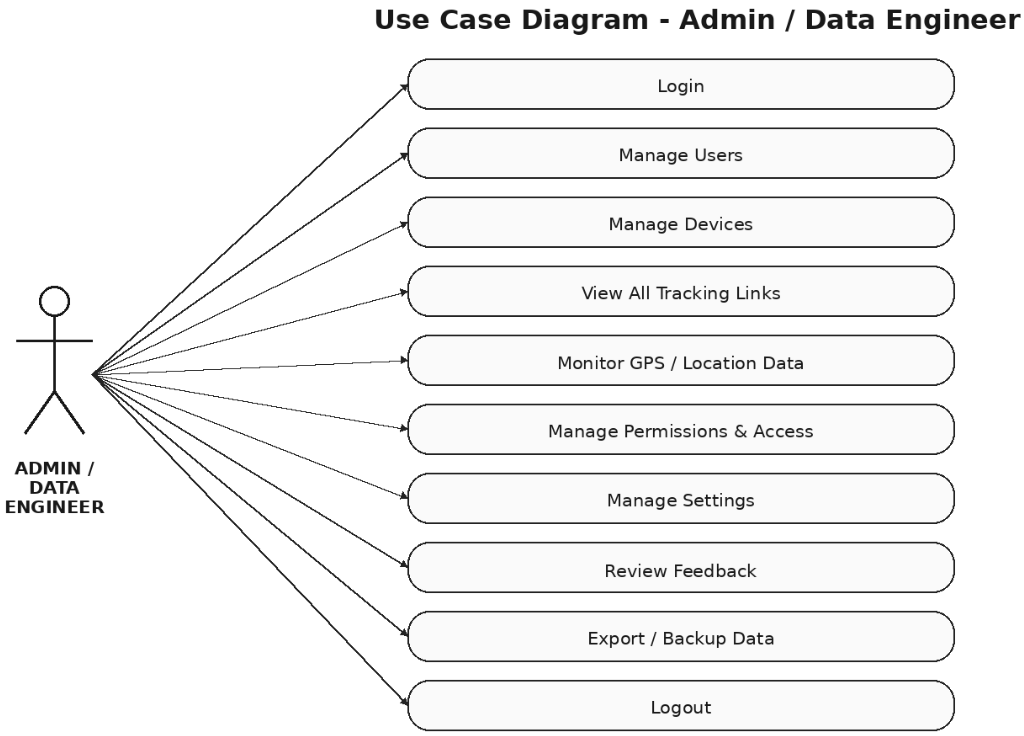

Figure 2: Use case diagram showing the interactions available to the Admin / Data Engineer actor.

Diagram interpretation: The administrator maintains the application's operational state by managing users, devices, tracking links, permissions, settings, feedback and data-related operations.

Figure 3 — Activity Diagram: TrackPoint Management Workflow

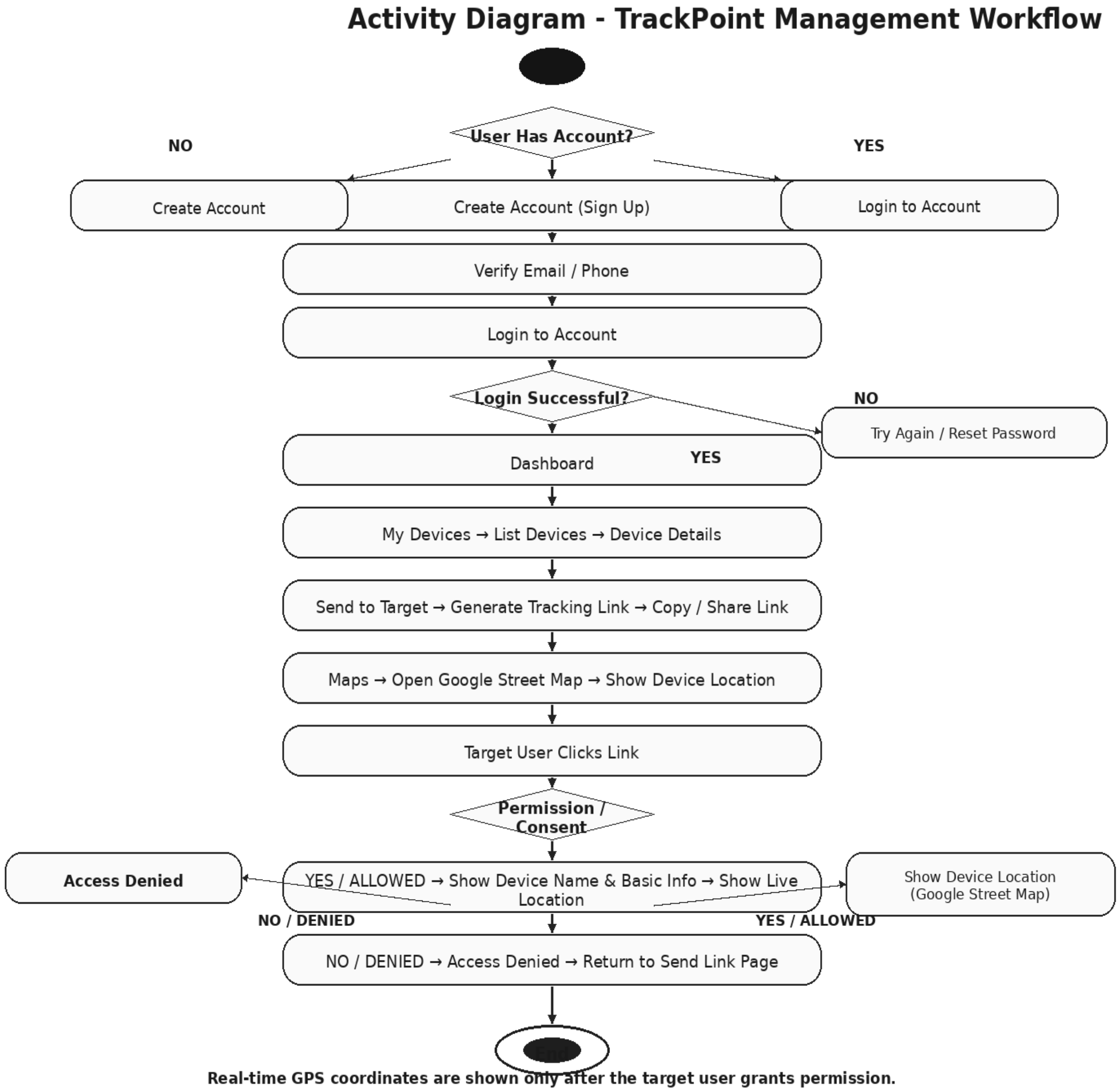

Figure 3: Activity diagram showing the end-to-end TrackPoint workflow, including account handling, dashboard access, device/link operations, map visualization and permission-controlled location disclosure.

Important branch: If the target user denies permission, access is denied. If permission is allowed, the system can display the device name/basic information and the permitted location on the map.

Figure 4 — Sequence Diagram: TrackPoint Location Request

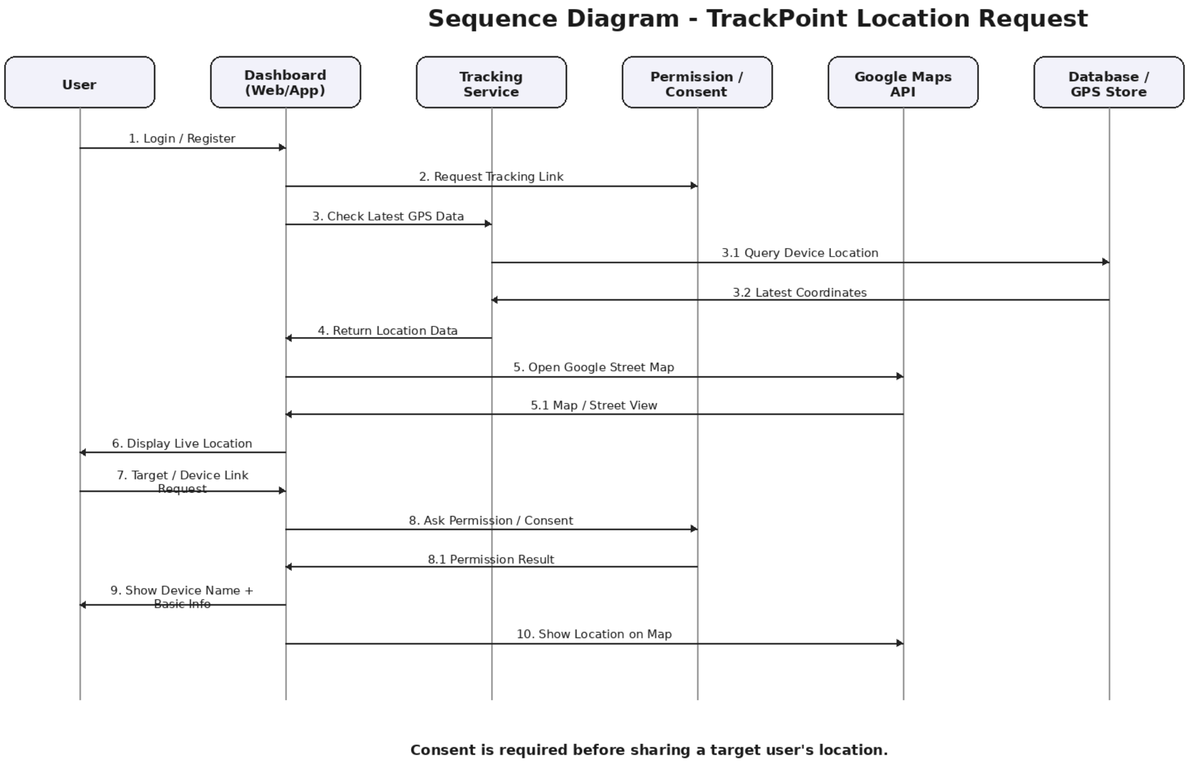

Figure 4: Sequence diagram showing interaction between the User, Dashboard, Tracking Service, Permission/Consent layer, Google Maps API and Database/GPS Store.

Sequence summary: The user requests tracking, the dashboard obtains the latest location information, permission is checked, the map service is invoked, and the permitted location is displayed to the user.

Complete system flow

User → Authentication → Dashboard → My Devices / Tracking Link → Target User → Permission / Consent → GPS Coordinates → Map API → Location Visualization → End.

Conclusion

TrackPoint Management provides a clear demonstration of real-time GPS tracking and map-based visualization while incorporating an explicit permission/consent step. Its user and administrator workflows make it suitable for controlled seminars, educational exercises and authorized ethical hacking demonstrations.